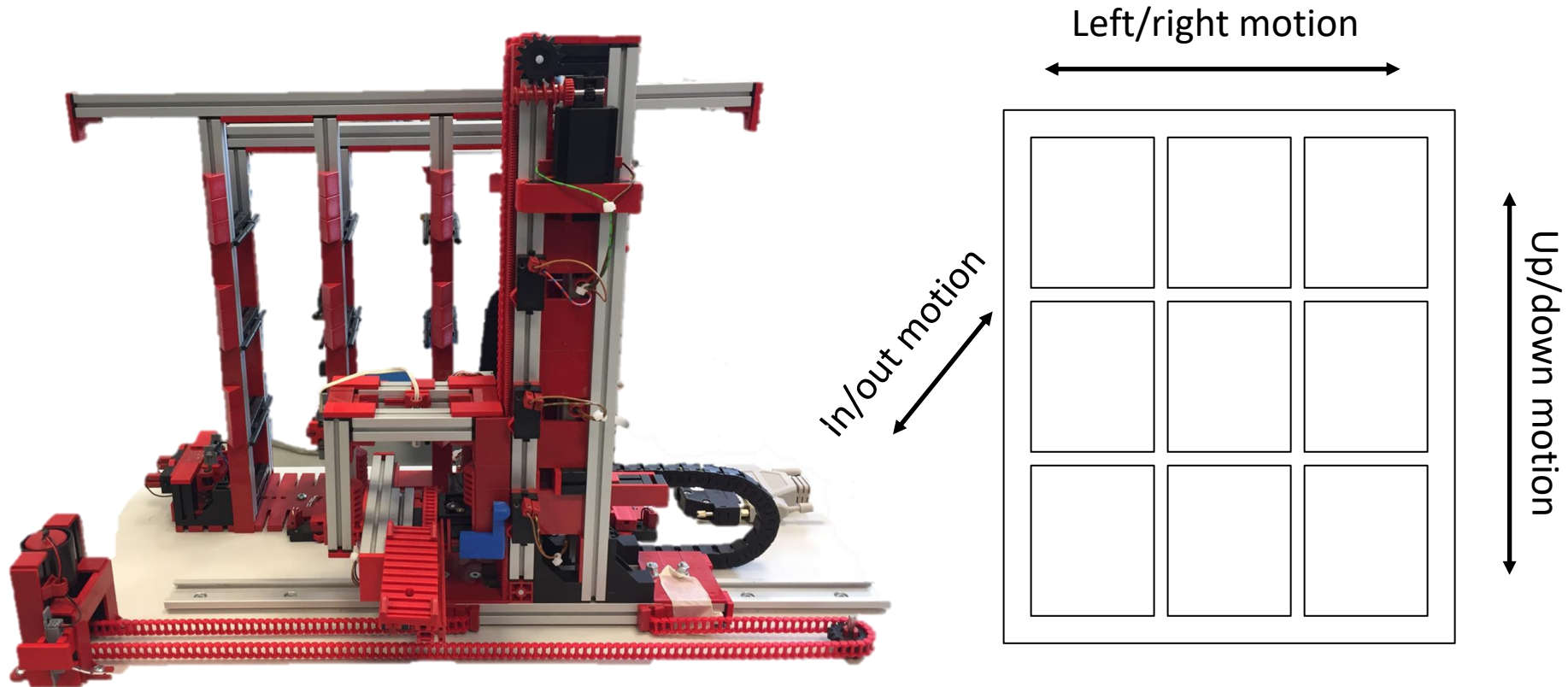


Available experiences:

1. Automation of an elevator plant
2. Control of electric drives and pneumatic actuators in an automated plant
- 3. Automation of a storage warehouse**
4. Control of a PMDC – PMDC system
5. Control of a PMSM – PMDC system
6. Control of linear vibrations
7. Industrial plant
8. Control of a linear pendulum
9. Control of a rotational pendulum I
10. Control of a rotational pendulum II
11. Control of a reaction pendulum
12. 2D ball balancing
13. Maglev
14. Magnetic bearing (Maglev 2)
15. Quanser Aero
16. Quadcopter

- Small scale model of a storage warehouse with 3x3 storage places
- Typical example of discrete event system to be controlled
- The control system is based on a PLC (ABB)
- The motion actuations are electromechanical (on/off)
- The control typology can be developed through increasing complexity levels.



Switches defining position and pallet presence (in storage places/
on the carrier)

Modelling and control approaches

- The system and the control evolve through discrete events.
- Possible modelling option:
 - finite state automaton (finite state machine)
 - Petri net
 - Flow diagram
 - SFC (Sequential Functional Chart aka GRAFCET)
 - other forms

- Definition and preliminary description of two basic control logics:
 - a) Simple storage/withdraw
 - b) Complex management of pallets with user defined sequences (and sequence optimisation)
- Accurate description of the sequences and modelling.
- Characterisation of the command outputs and of the sensor and command signals (level and car calls)
- Implementation of the two logics a) and b) on the PLC (ladder diagram/ structure text /any of IEC 61131)
- Addition of all the safety features